

KMT-2026-BLG-0521 / OGLE-2026-BLG-0058

The v5 reduction — restricted set, freely fitted, Gaia applied afterwards

Prepared 2026-09-09. Solution: `~/KMTdata/Results/v5/IFfinal_<field>.mat` (variable IFsys; the logical Frame in the same file marks the calibration stars). This document describes **only** v5. The reference decade reduction is in `KMT260058_astrometry_report.pdf`; the Gaia-pinned variant and the v4/v5 comparison are in `KMT260058_v4_report.pdf`.

1. What v5 is, in one paragraph

The analysis is restricted to a clean calibration set — bright, isolated, and astrometrically well behaved in Gaia — plus the target and three comparison stars at the target's own magnitude. Every one of those stars is **freely fitted** and every one takes part in determining the per-epoch frame at every step. Gaia is used only to *select* the set, and then again at the very end, as a post-hoc tie that fixes the frame's gauge. Nothing is held at a catalogue value during the fit.

This is the counterpart to v4, which uses the identical source set but pins each calibration star's position and proper motion to Gaia while fitting.

2. Object selection, step by step

Both fields, from the raw matched-source file down to what is actually fitted:

step	BLG41	BLG01
raw sources in the MSc file	1177	1187
survive the standard quality cuts	594	621
have an OGLE I magnitude	540	478
in $14 < I < 17$	292	253
... and isolated: no $I < 18$ companion within 2.0 arcsec	144	127
... and matched to a Gaia DR3 source	144	115
... and that source has a RUWE	142	111
... and RUWE < 1.4 → the calibration set	126	96
+ the target + 3 comparison stars	+4	+4
= analysis set actually fitted	130	100

Notes on the individual cuts:

- The **quality cuts** are the pipeline's standard ones and are not part of this design. They remove sources detected in a median of 10% of epochs, against 80% for those retained; every source found in more than half the epochs survives. The restriction imposed by v5 is therefore **130 of 594**, not 130 of 1177.
- The **magnitude window** costs about half the OGLE-matched sources. It is the single largest cut after the

standard ones.

- The **isolation cut** costs another half: 292 → 144 on BLG41. The companion list is the OGLE catalogue, which resolves about six times more sources than KMT detects, so it finds close pairs that our own source list has merged.
- The **RUWE cut** removes 11% (BLG41) and 14% (BLG01) of what reaches it, consistent with the 87% pass rate for $G < 17$ across the whole cone. Crowding is not biasing it heavily.
- BLG01 loses 12 sources at the Gaia-match step where BLG41 loses none. This follows its poorer OGLE coverage, 478 of 621 against 540 of 594.
- **The target is not in the calibration set** — at $I = 18.13$ it falls outside the magnitude window — and is added explicitly.

3. Gaia

Gaia enters twice, and in neither case during the fit.

Selecting the set. RUWE is **not** in the local catsHTM DR3 table, which carries `astrometric_chi2_al` and `astrometric_n_good_obs_al` but neither RUWE nor the `u0(G, BP-RP)` table needed to normalise UWE into it. The true values were queried from the Gaia archive — 9464 sources in a 3 arcmin cone, cached in `~/KMTdata/GaiaRef/`. This mattered: at $G < 17$, **87%** pass RUWE < 1.4 , where the unnormalised UWE proxy predicted 74%.

Fixing the gauge, afterwards. A free fit determines relative astrometry only: applying any global affine to all source positions, and compensating in the per-epoch transformations, leaves every residual unchanged. Six parameters are therefore pure gauge, and the same freedom appears one time-derivative up in the proper motions as a constant plus a linear gradient across the field. The tie removes it. The gauge found was:

	$\mu_\alpha \cos \delta$	μ_δ	calibrators
BLG41	-0.219	+4.758	130
BLG01	+2.834	+3.983	100

mas/yr. With it removed, the seven plotted stars agree between the two fields to 0.25 to 0.78 mas/yr in RA and 0.02 to 0.87 in Dec.

A caution on the tie method. `ml.util.gaiaTie` fits a 2 by 2 matrix from a proper-motion regression. On these small sets it is poorly determined: it returned rotations of **146.34 deg (BLG41) and 108.65 (BLG01)**, 38 deg apart, where the direct positional Gaia-to-pixel map puts the two cut-outs 0.126 deg apart. Its linear part is fitting noise. The gauge above was therefore taken the other way round: the linear part from the positional map, which is solid, and only the constant from the proper-motion comparison. The two methods differ by 0.2 to 0.6 mas/yr on the target, which is a systematic worth carrying.

4. The chain

Two steps, on the 130 / 100 source subset:

1. **Full-decade SysRem correction.** 2 components on the astrometric residuals, computed once over the whole run and reused. 2. **Final solution**, reusing that correction, with the pixel-phase term on, 15 weighted iterations before SysRem and 6 after.

The joint proper-motion step and the per-season fits of the reference reduction are not used: with so few, such well-measured sources the motions are solved directly in the global fit.

	BLG41	BLG01
step 1	16.3 min	14.8 min
step 2	30.0 min	28.0 min
convergence over the calibration stars, last 3 iterations	0.0041 mas	0.0002 mas

5. Results

	BLG41 $\Delta X / \Delta Y$	BLG01 $\Delta X / \Delta Y$
calibration stars (126 / 96)	4.87 / 4.81	4.77 / 4.77
the target	25.61 / 22.27	28.45 / 23.15

mas. **4.8 mas is the best frame-star astrometry this project has produced** — better than the reference reduction's bright stars (6.41 / 6.88 and 6.48 / 7.01) and better than the same objects pinned to Gaia in v4 (6.13 / 5.20 and 5.30 / 5.42). Some of that advantage is self-fulfilling: a free fit adjusts four parameters per star and so absorbs part of each star's own systematics.

The target is **not** improved. The reference reduction measures it at 24.62 / 21.00 and 25.48 / 21.78, better on every axis in both fields. Restricting the analysis set costs 4 to 12% on the target while improving the calibration stars, because the per-epoch frame is now built from 126 stars instead of ~600 weighted sources.

Target proper motion, absolute, gauge removed:

	$\mu_{\alpha \cos \delta}$	μ_{δ}
BLG41	-1.80	-6.20
BLG01	-2.13	-6.51

mas/yr; the two fields differ by 0.33 / 0.31. Using `gaiaTie`'s own matrix instead gives -1.978 / -6.757 and -1.906 / -6.880, differing by 0.072 / 0.123. The honest statement is that the fields agree to **0.07 to 0.36 in RA and 0.12 to 0.29 in Dec**, the spread being the choice of tie method.

6. Comparison stars

Six per field, all freely fitted, all plotted in the same style as the target:

tag	I	distance from the target	decade residual RMS, BLG41 / BLG01
cal_d24	16.68	24 pix	4.07 / 4.14
cal_d30	16.63	30 pix	6.73 / 6.83
cal_d43	16.94	43 pix	4.48 / 4.60
out_d04	18.14	4.5 pix	28.61 / 32.37
out_d21	17.91	21 pix	17.36 / 17.54
out_d48	18.05	48 pix	19.12 / 18.59

Coordinates. Pixel positions are per field, since the two cut-outs are offset by about 1.1 pixels; sky positions are derived by inverting each field's own Gaia-to-pixel map, and the two fields agree on them to **2 to 9 mas**, which is an independent check on both maps.

star	I_OGLE	dist [pix]	BLG41 pixel	BLG01 pixel	RA (J2000)	Dec (J2000)
target	18.13	0.0	150.4, 150.4	151.6, 150.2	17:52:38.085	−31:47:36.21
cal_d24	16.68	24.5	161.3, 172.2	162.4, 172.1	17:52:37.746	−31:47:27.53
cal_d30	16.63	30.5	120.4, 154.6	121.4, 154.5	17:52:39.023	−31:47:34.60
cal_d43	16.94	42.9	110.9, 166.6	111.9, 166.5	17:52:39.318	−31:47:29.89
out_d04	18.14	4.5	146.4, 148.3	147.6, 148.0	17:52:38.210	−31:47:37.06
out_d21	17.91	21.3	155.3, 171.1	156.4, 170.9	17:52:37.935	−31:47:28.01
out_d48	18.05	48.5	198.6, 145.4	199.8, 145.0	17:52:36.585	−31:47:38.10

The target's recovered position, 17:52:38.085 −31:47:36.21, sits 0.11 arcsec from the OGLE EWS coordinates the pipeline was pointed at, which is the accuracy of the centroid plus the map, not a disagreement.

mas in X. The three cal_ stars are members of the calibration set, chosen from those present in **both** fields so the same physical object carries the same name everywhere. The three out_ stars sit at the target's own magnitude and are deliberately outside the set, to give something comparable to the target.

out_d04 is not a usable comparison object. It lies 4.5 pixels from the target under a seeing FWHM near 7 pixels, so it shares most of its light with it. Its residual RMS is 29 to 32 mas against 17 to 19 for the other two stars of the same brightness, in every season and not only during the event; and its 2026 season is displaced by 40 to 58 mas as the target brightens. Its CSV carries a warning in the header. out_d21 and out_d48 are the honest controls, and their errors bracket the target's own 25 to 27 mas — the check that the target is not anomalously noisy for what it is.

7. Files

file	content
report_v5_RMS.png	residual RMS against OGLE I, per axis and field, for this run
report_v5_motion_<field>_target.png	the target, sidereal-month bins, 2x2
report_v5_motion_<field>_cal_d{24,30,43}.png	the three calibration comparison stars
report_v5_motion_<field>_out_d{04,21,48}.png	the three I ~ 18 comparison stars
source_motion_v5_<field>.csv	the target, per epoch
source_motion_v5_<field>_<tag>.csv	each comparison star, per epoch

CSV columns are JD, X_mas, Y_mas, errX_decade, errY_decade, errX_season, errY_season, season. Positions are relative to that star's own mean, with **proper motion not removed**. err_decade is that star's own

residual RMS over the whole run and `err_season` the same within each season; note this differs from the reference reduction's CSV, where the decade column held a field-star population estimate — v5 keeps no field stars at $I \sim 18$, so that curve cannot be formed here.

In every motion figure the columns are the two axes, **X left and Y right**; the top row keeps the proper motion and the bottom row removes it. The legend quotes the motion along the pixel axis and the heading quotes it on the sky with the gauge removed. **Pixel X runs opposite to RA**, so those two carry opposite signs in X by construction.

Figures

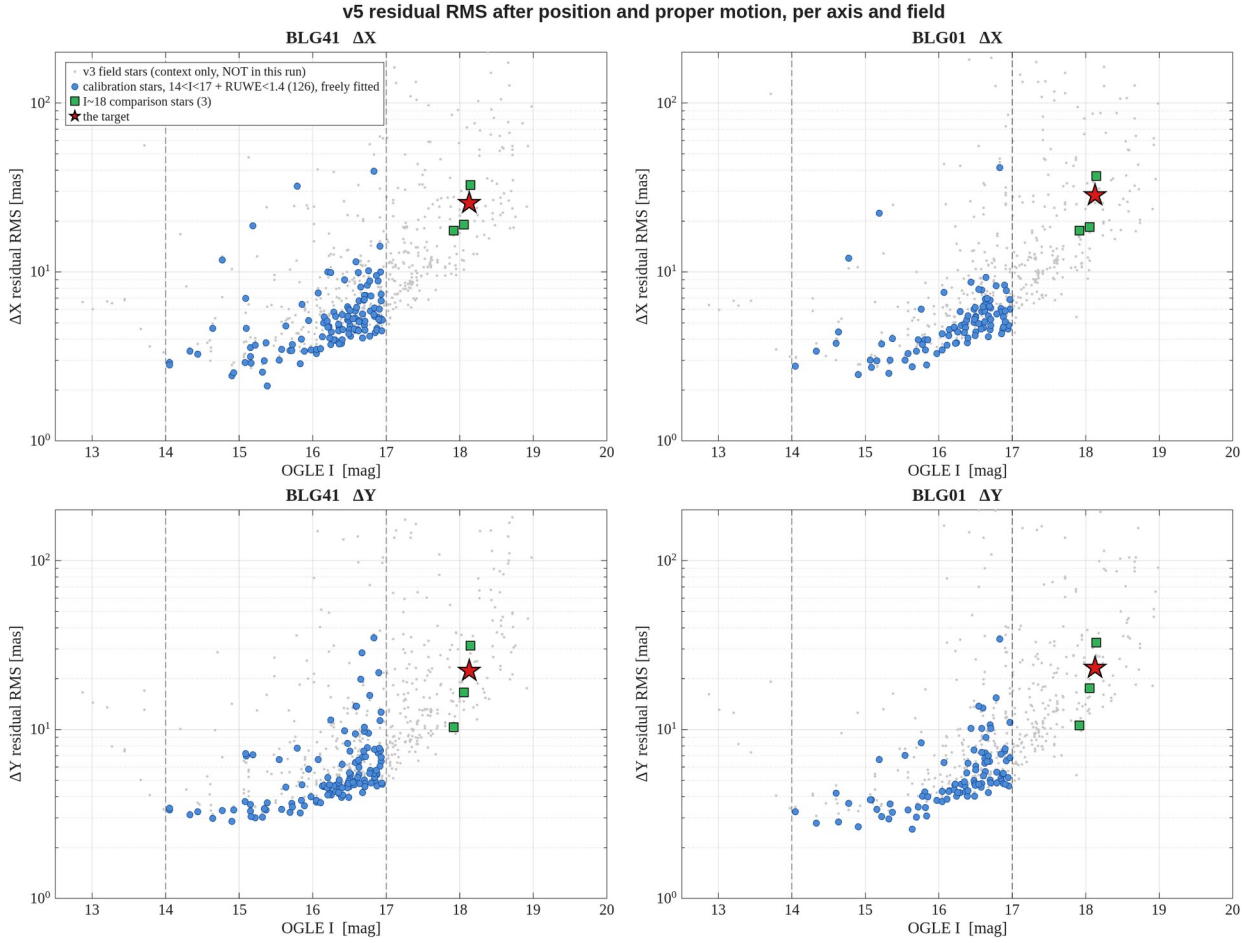


Figure 1. Residual RMS against the OGLE catalogue magnitude, per axis and per field, for the v5 run, after position and proper motion have been removed. Blue: the 126 / 96 calibration stars, $14 < I < 17$ with $\text{RUWE} < 1.4$, **freely fitted**. Green: the three $I \sim 18$ comparison stars. Red: the target. Grey: field stars from the reference reduction, shown for context only — **they are not part of this run**. Dashed lines mark the 14 and 17 mag limits.

report_v5_RMS.png

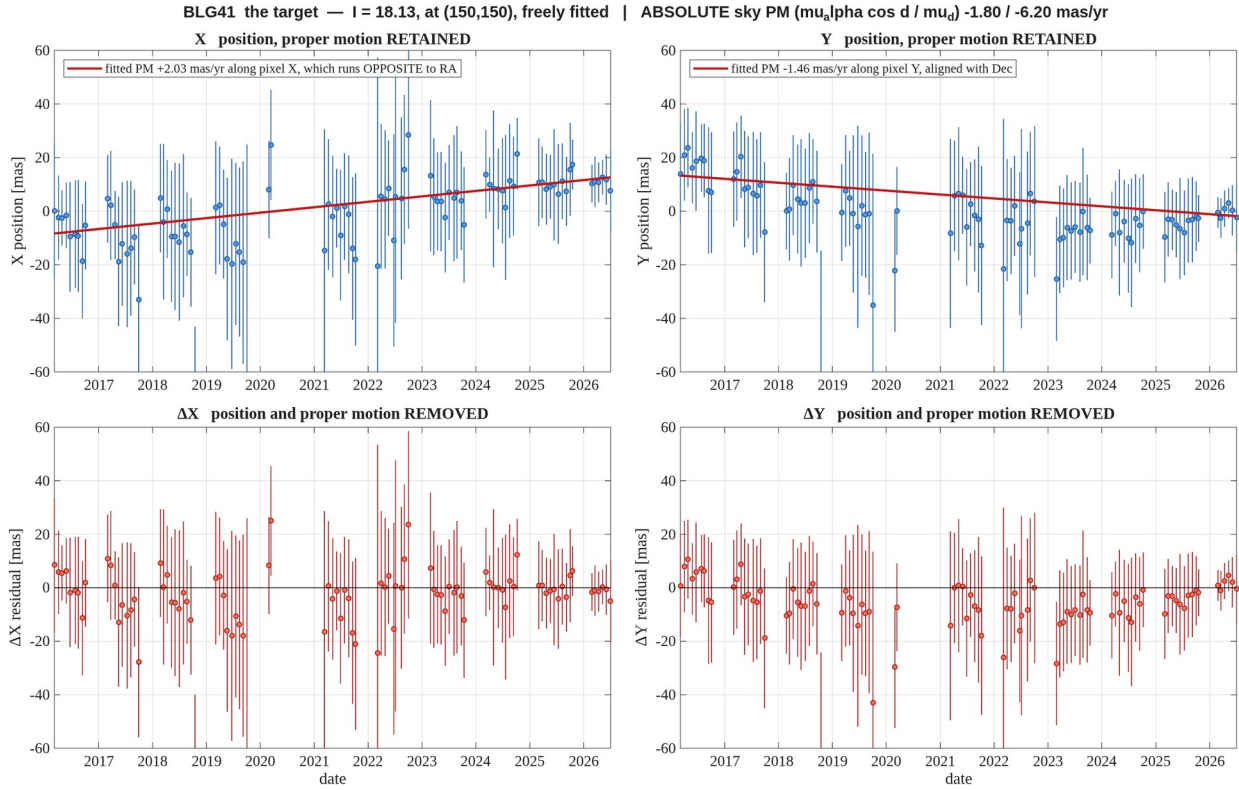


Figure 2. **BLG41** — the target, $l = 18.13$. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG41_target.png

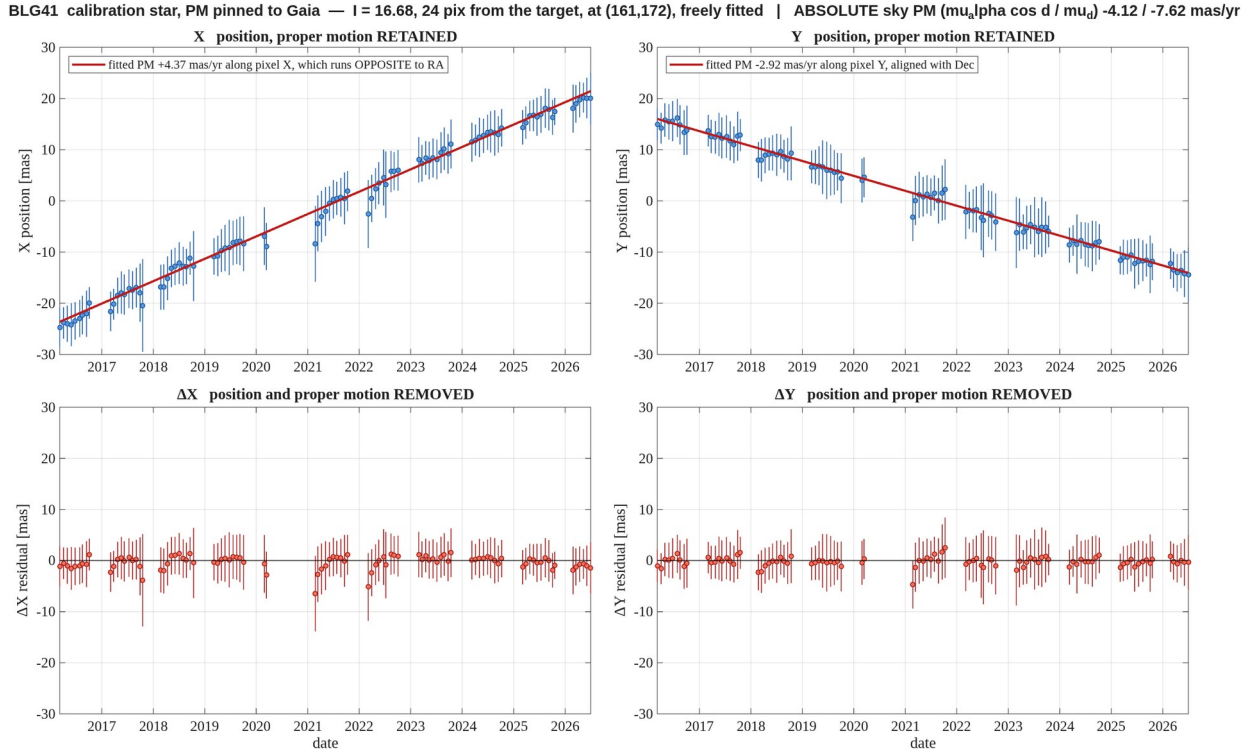


Figure 3. **BLG41** — calibration star, $l = 16.68$, 24 pix from the target. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG41_cal_d24.png

BLG41 calibration star, PM pinned to Gaia — $l = 16.63$, 30 pix from the target, at (120,155), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \lvert \cos d / \mu_\delta$) $+2.20 / +1.95$ mas/yr

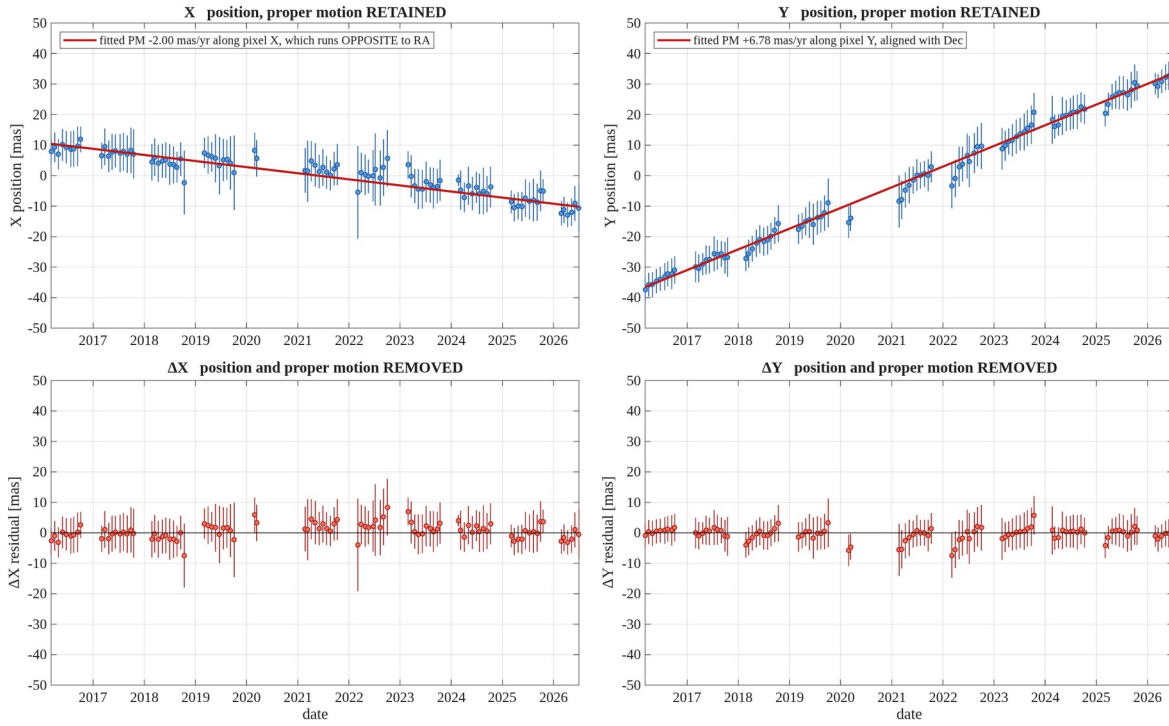


Figure 4. **BLG41** — calibration star, $l = 16.63$, 30 pix from the target. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG41_cal_d30.png

BLG41 calibration star, PM pinned to Gaia — $l = 16.94$, 43 pix from the target, at (111,167), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \lvert \cos d / \mu_\delta$) $-0.28 / -5.96$ mas/yr

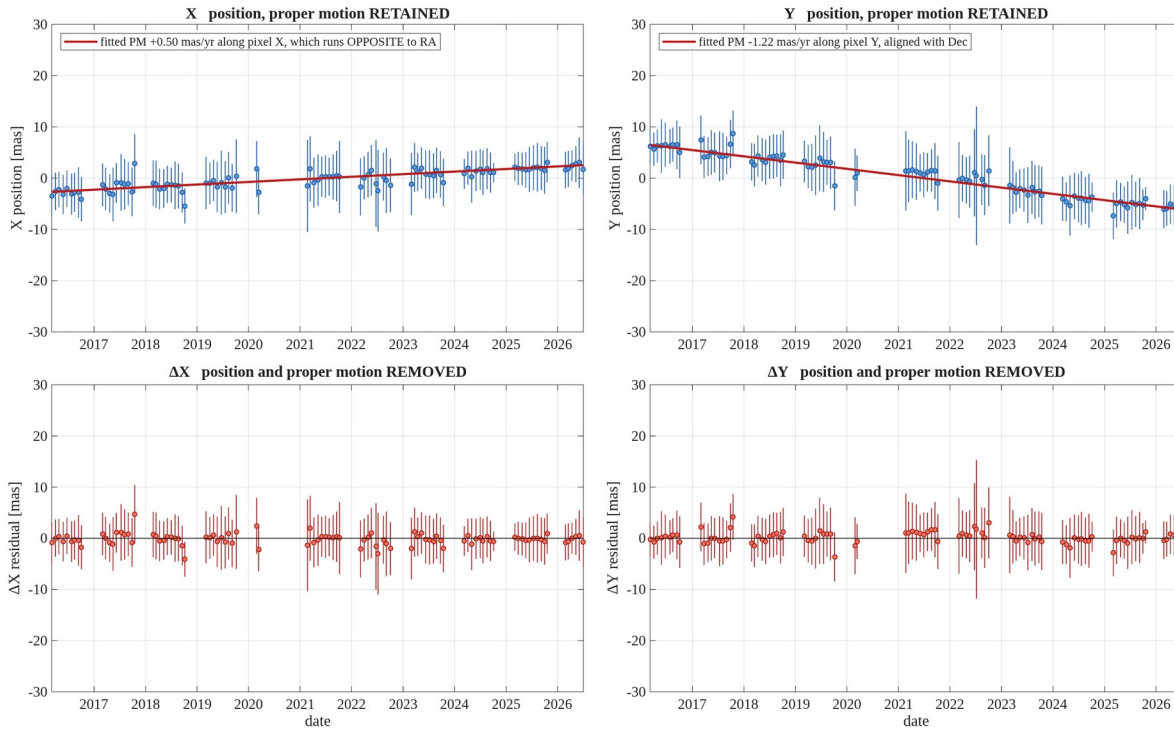


Figure 5. **BLG41** — calibration star, $l = 16.94$, 43 pix from the target. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG41_cal_d43.png

BLG41 comparison star outside the set, PM fitted — $l = 18.14$, 5 pix from the target, at (146,148), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos \delta / \mu_\delta$) $-0.17 / -4.55$ mas/y

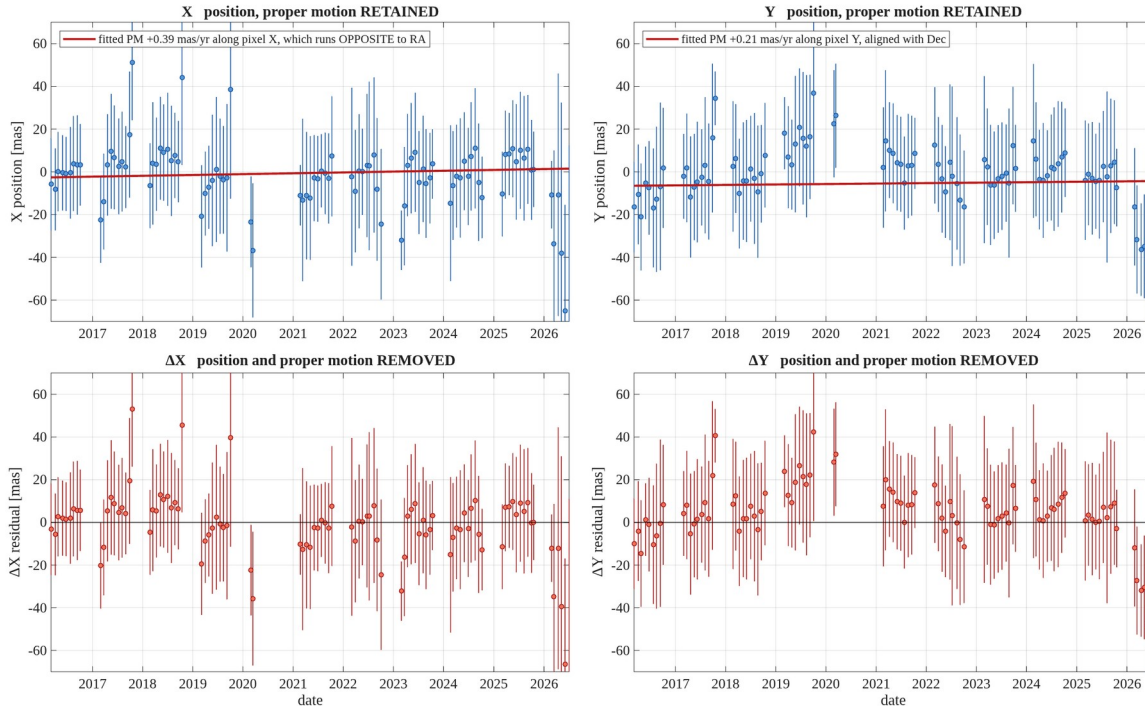


Figure 6. **BLG41** — comparison star $l = 18.14$, 4.5 pix away — **blended with the target**, see section 6. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X. report_v5_motion_BLG41_out_d04.png

LG41 comparison star outside the set, PM fitted — $l = 17.91$, 21 pix from the target, at (155,171), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos \delta / \mu_\delta$) $-7.71 / -9.55$ mas/y

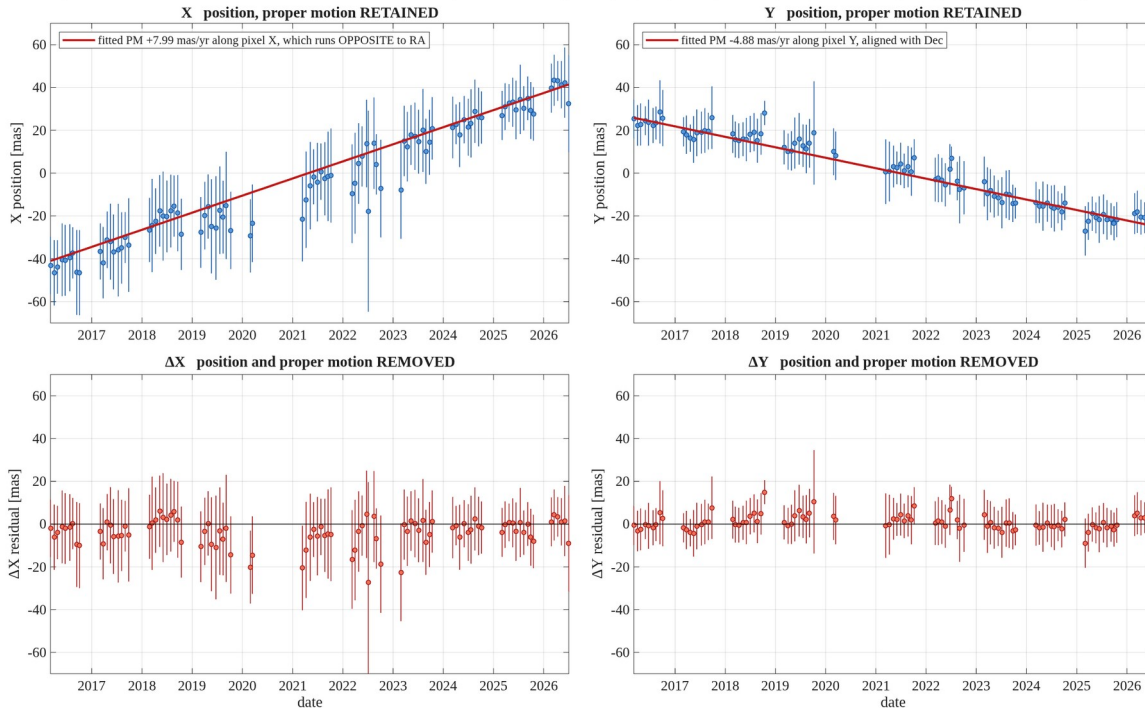


Figure 7. **BLG41** — comparison star $l = 17.91$, 21 pix away. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X. report_v5_motion_BLG41_out_d21.png

BLG41 comparison star outside the set, PM fitted — $l = 18.05$, 48 pix from the target, at (199,145), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos d / \mu_\delta$) +3.08 / -7.56 mas/

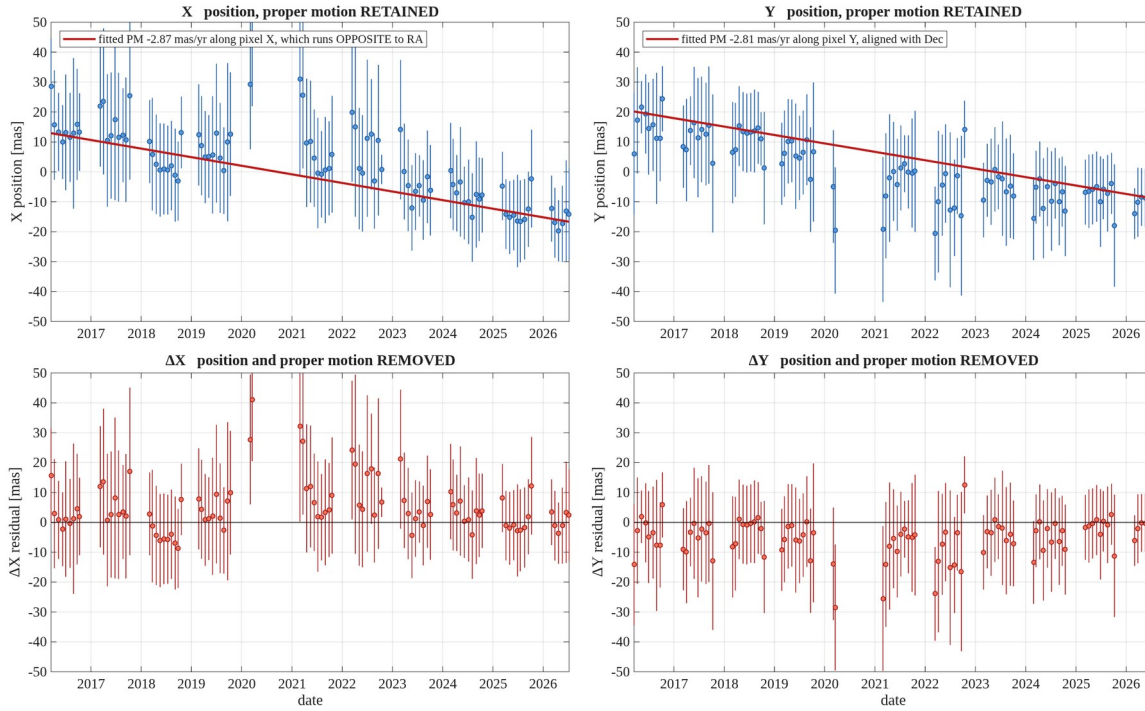


Figure 8. **BLG41** — comparison star $l = 18.05$, 48 pix away. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG41_out_d48.png

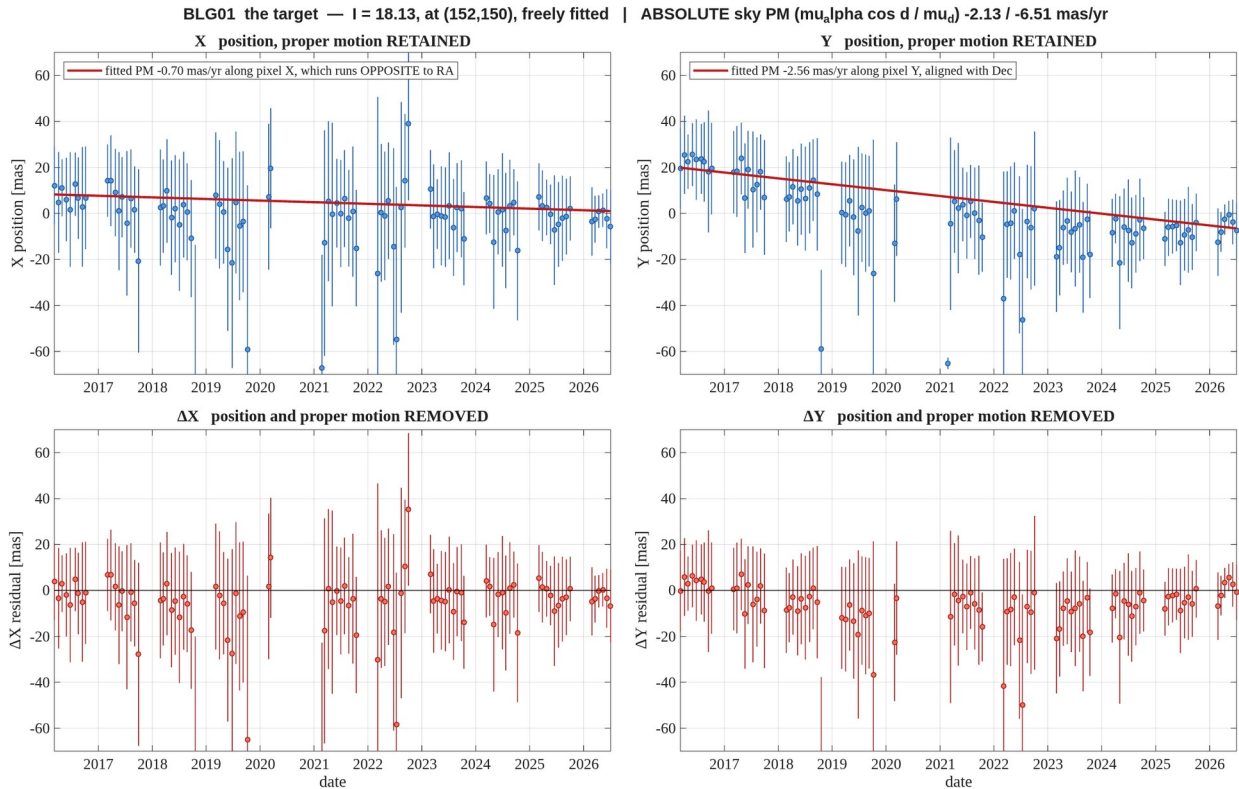


Figure 9. **BLG01** — the target, $l = 18.13$. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG01_target.png

BLG01 calibration star, PM pinned to Gaia — $l = 16.68$, 24 pix from the target, at (162,172), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos d / \mu_\delta$) -4.59 / -7.73 mas/yr

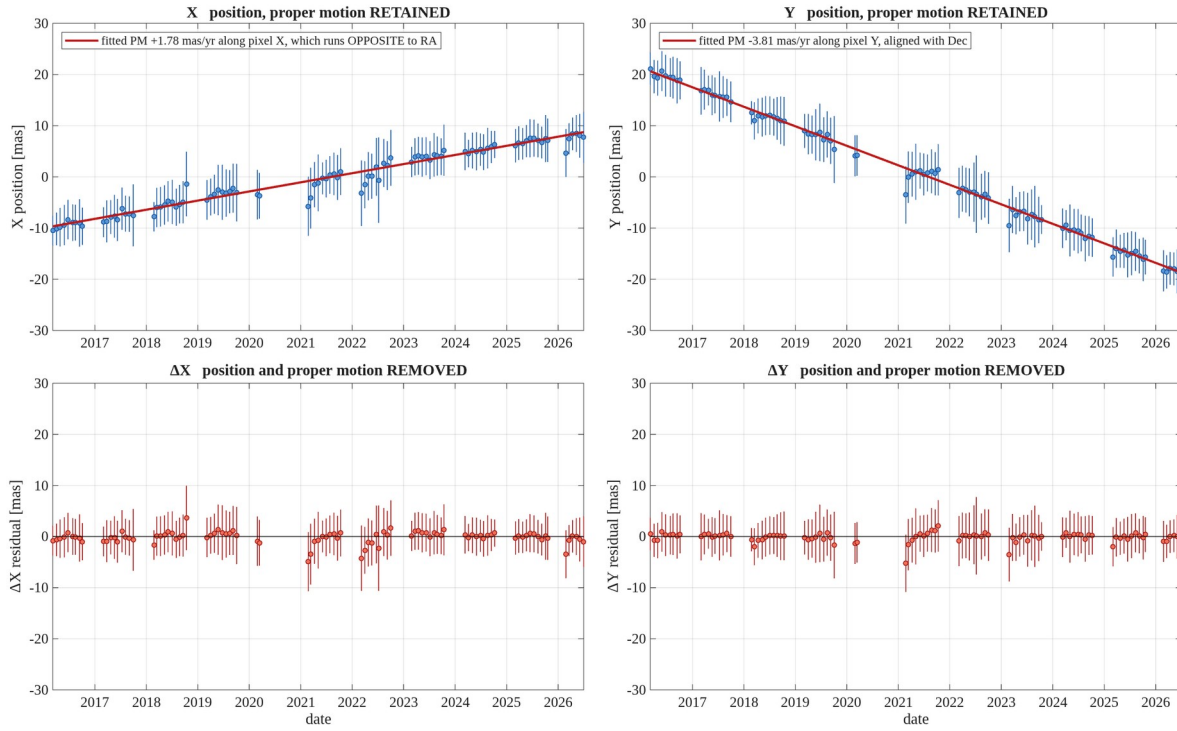


Figure 10. **BLG01** — calibration star, $l = 16.68$, 24 pix from the target. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG01_cal_d24.png

BLG01 calibration star, PM pinned to Gaia — $l = 16.63$, 30 pix from the target, at (121,154), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos d / \mu_\delta$) +1.58 / +1.97 mas/yr

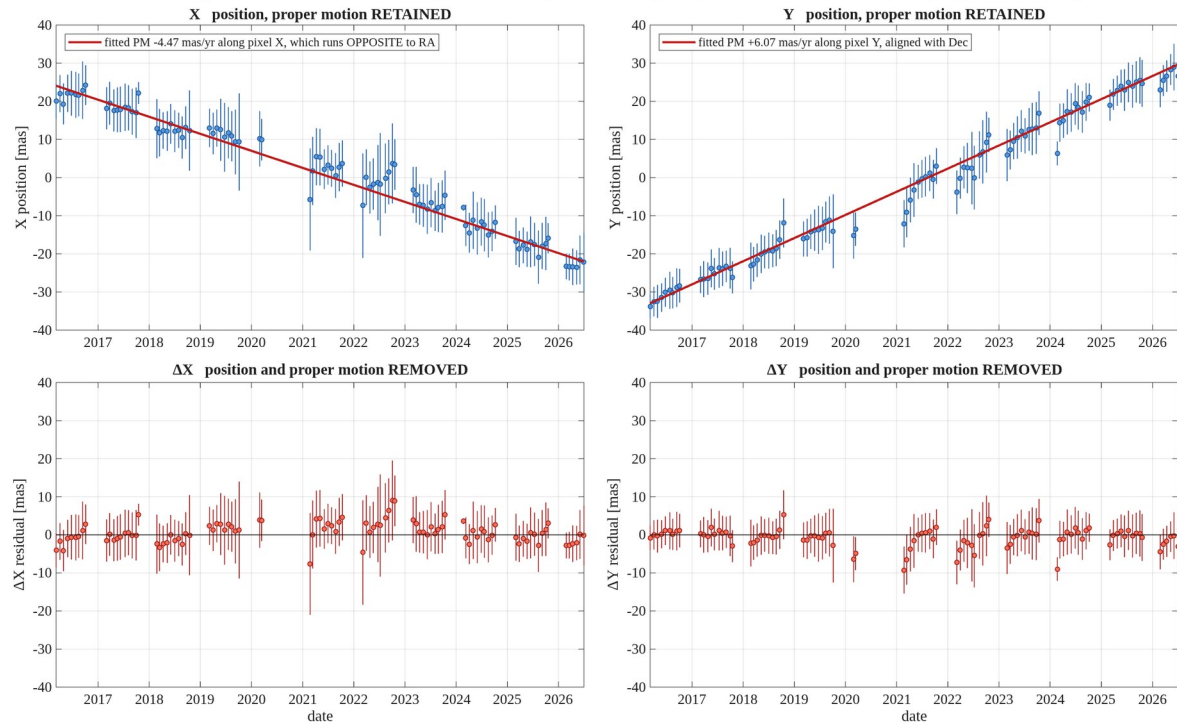


Figure 11. **BLG01** — calibration star, $l = 16.63$, 30 pix from the target. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG01_cal_d30.png

BLG01 calibration star, PM pinned to Gaia — $l = 16.94$, 43 pix from the target, at (112,166), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos d / \mu_\delta$) -1.00 / -5.99 mas/yr

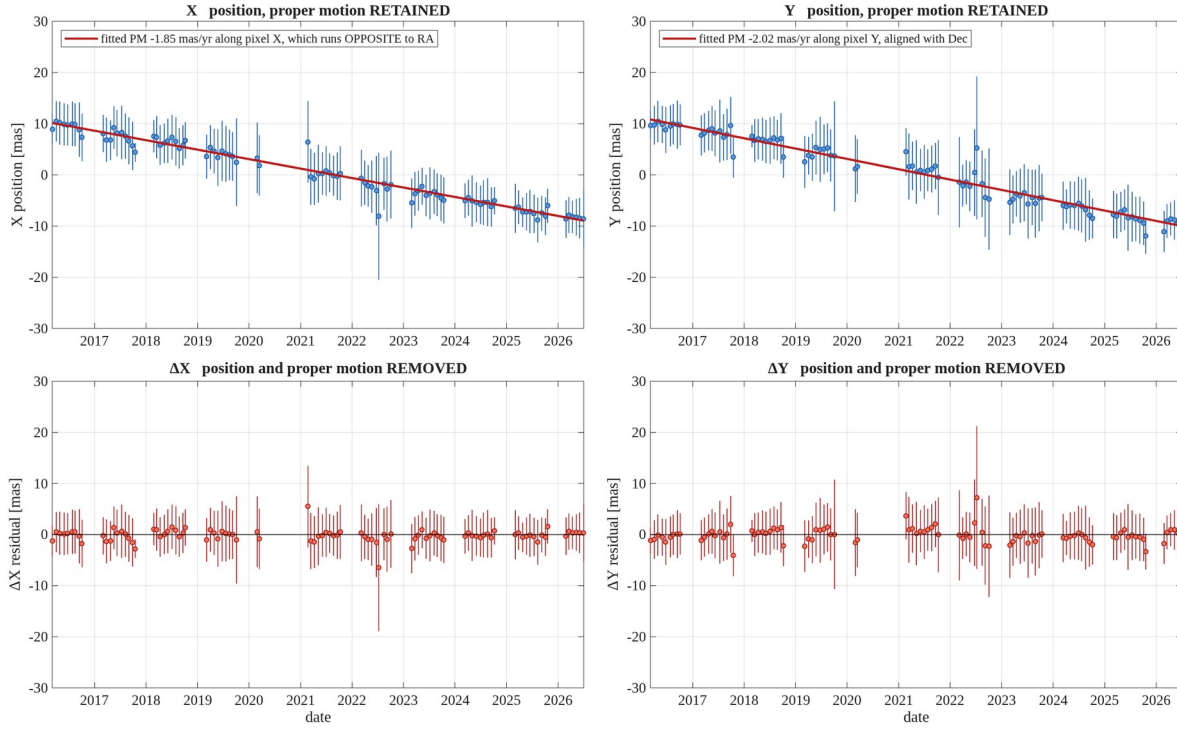


Figure 12. **BLG01** — calibration star, $l = 16.94$, 43 pix from the target. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG01_cal_d43.png

BLG01 comparison star outside the set, PM fitted — $l = 18.14$, 5 pix from the target, at (148,148), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos d / \mu_\delta$) -0.71 / -5.42 mas/yr

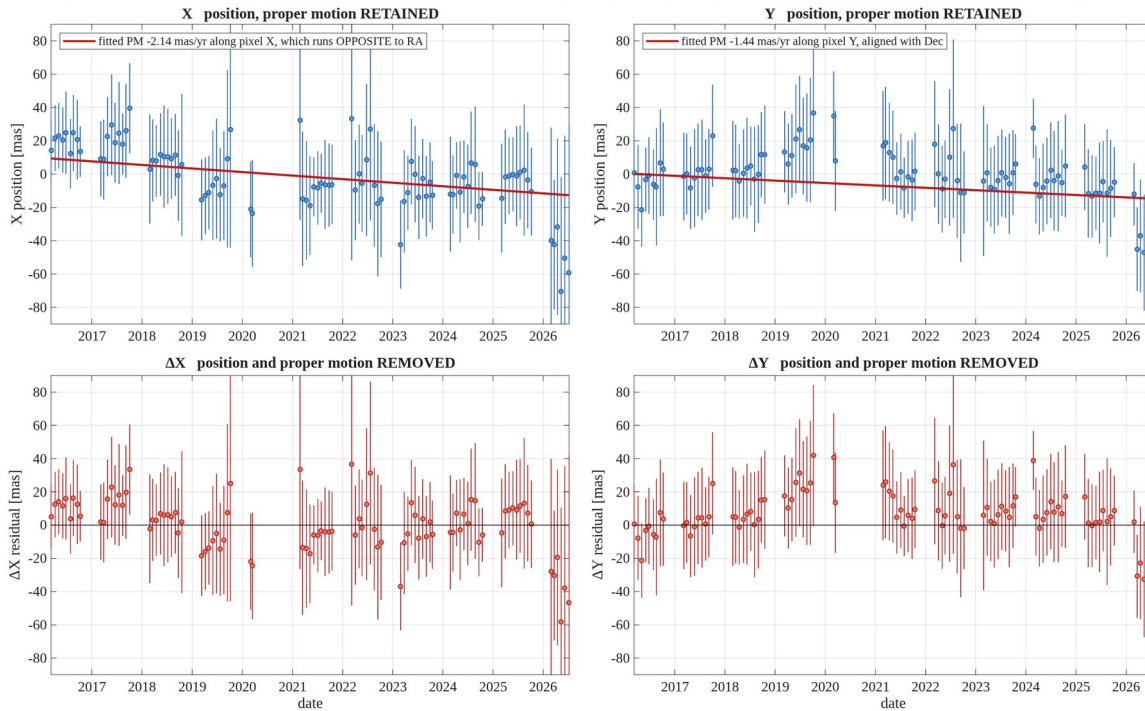


Figure 13. **BLG01** — comparison star $l = 18.14$, 4.5 pix away — **blended with the target**, see section 6. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG01_out_d04.png

LG01 comparison star outside the set, PM fitted — $l = 17.91, 21$ pix from the target, at (156,171), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos d / \mu_\delta$) $-8.49 / -9.74$ mas/

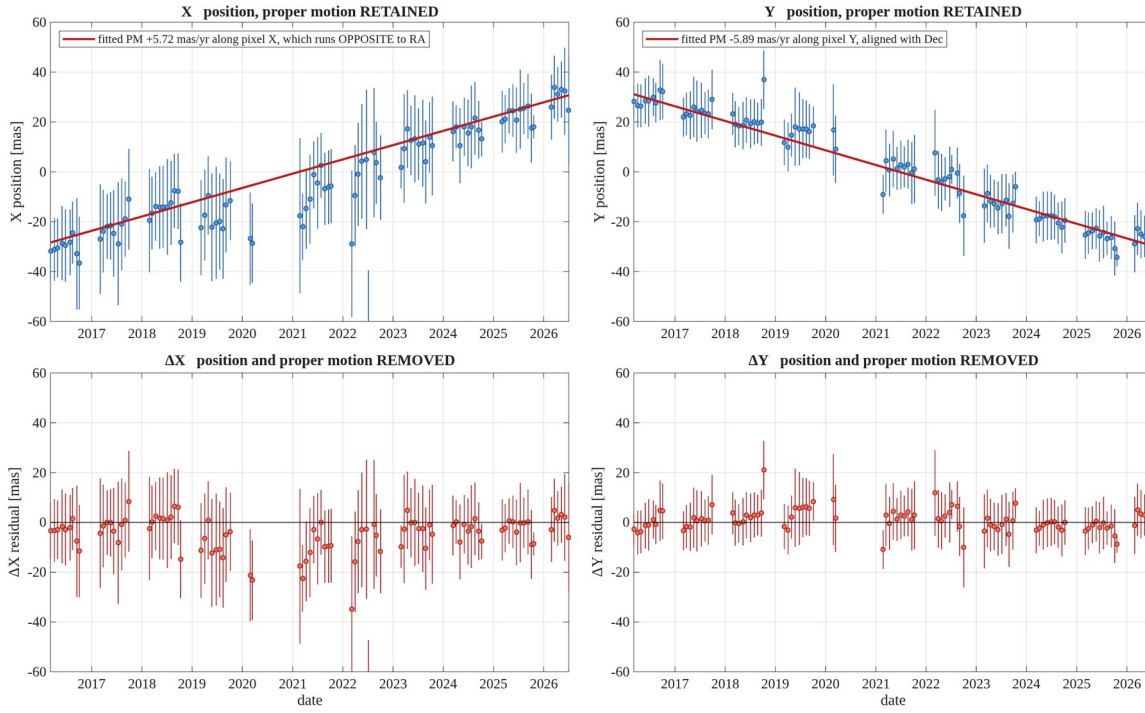


Figure 14. **BLG01** — comparison star $l = 17.91, 21$ pix away. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG01_out_d21.png

LG01 comparison star outside the set, PM fitted — $l = 18.05, 49$ pix from the target, at (200,145), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos d / \mu_\delta$) $+2.83 / -7.32$ mas/

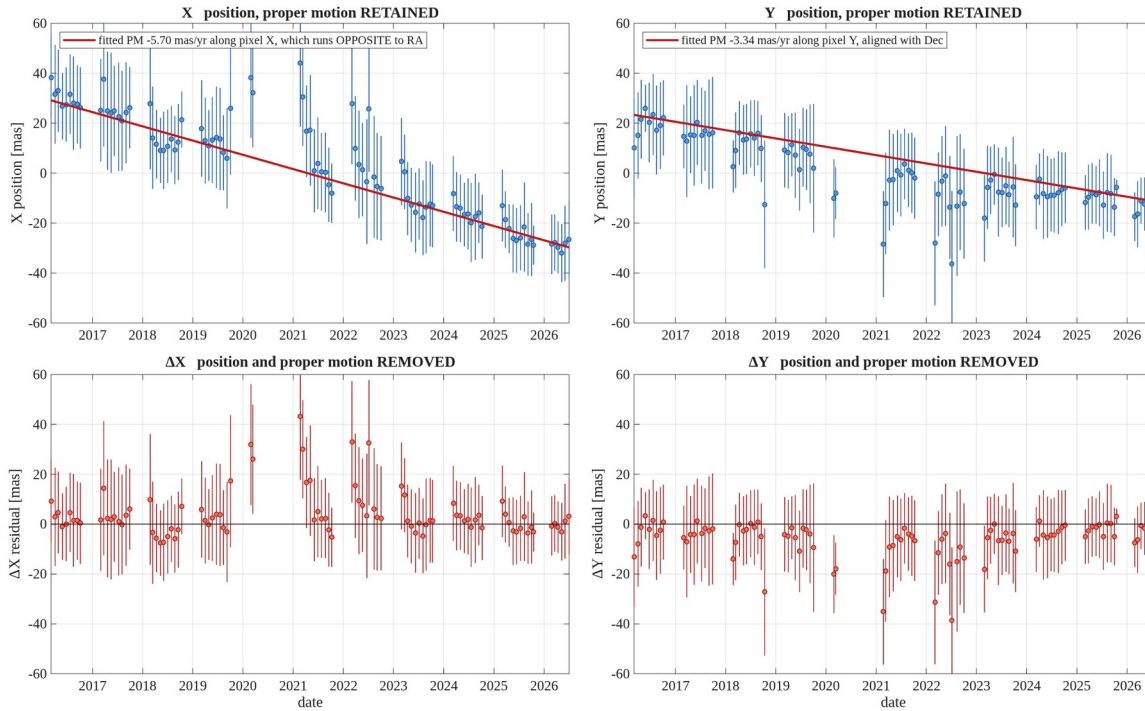


Figure 15. **BLG01** — comparison star $l = 18.05, 48$ pix away. Every star here is freely fitted; nothing is pinned. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

report_v5_motion_BLG01_out_d48.png